TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 17, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Analyze the band structure diagram of a topological insulator shown below. Identify where band inversion occurs and explain how this leads to a non-trivial topology in the system.

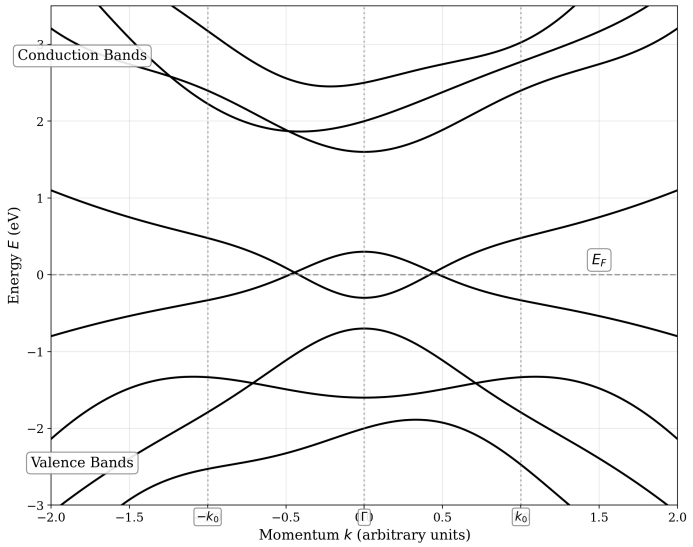

Fig. 1. Band structure of a topological insulator showing the conduction and valence bands with band inversion highlighted. The diagram should include the energy axis and momentum axis, with the band inversion region clearly marked to illustrate the non-trivial topology. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Brillouin • band inversion The diagram will FAIL review if any of these appear.

Question 2 10.0 points

Consider the surface state dispersion diagram shown below for a 3D topological insulator. Analyze the Dirac cone structure and explain the significance of protected surface states as a consequence of the bulk-boundary correspondence.

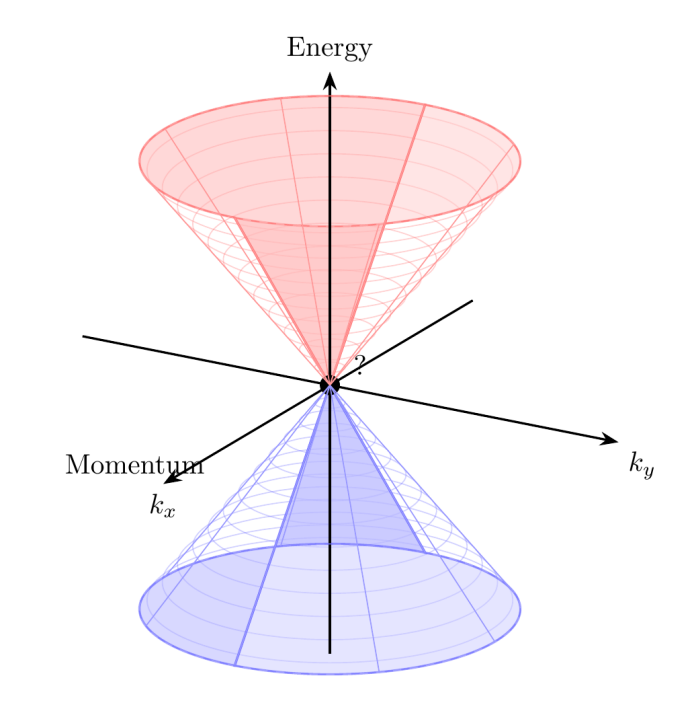

Fig. 2. Dispersion relation showing Dirac cone on the surface of a 3D topological insulator. The diagram should depict the linear Dirac cone structure with the two cones touching at a single point. The axes should be labeled with energy (E) on the vertical axis and momentum (k) on the horizontal axes, indicating the surface state dispersion. Highlight the Dirac point where the cones meet, representing the protected surface states due to bulk-boundary correspondence. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • dirac cone • linear dispersion The diagram will FAIL review if any of these appear.

Question 3

10.0 points

Explain how the Z_2 topological invariant is used to determine the topological phase of a quantum spin Hall insulator in the diagram below. Provide a brief outline of the calculation method in k-space.

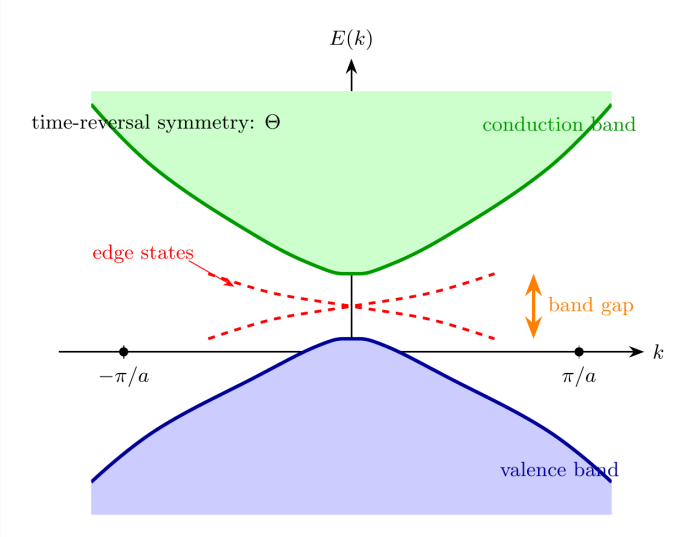

Fig. 3. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Brillouin; The diagram contains labels that directly answer parts of the question, specifically "Z2 invariant calculated at TRIM" and "band inversion region", which are key concepts the student is expected to explain.; The diagram contains the label "TRIM" pointing to the special momentum points on the k-axis, which is a key concept the student is supposed to identify and explain as part of the calculation method for the Z2 invariant.. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: Brillouin; The diagram contains labels that directly answer parts of the question, specifically "Z2 invariant calculated at TRIM" and "band inversion region", which are key concepts the student is expected to explain.. A band structure diagram of a quantum spin Hall insulator. The diagram shows the energy E(k) versus momentum k. It includes a 'conduction band' (top, shaded green) and a 'valence band' (bottom, shaded blue). The bands show an inversion near the Γ point (k=0). Dashed red lines labeled 'edge states' cross the band gap, connecting the valence and conduction bands. The k-axis is labeled with time-reversal invariant momenta (TRIM) points: Γ, -π/a, and π/a. An annotation for 'time-reversal symmetry: Θ' is present. A 'band gap' is indicated by a vertical orange arrow. Do NOT include the labels "Z2 invariant calculated at TRIM" or "band inversion region". IMPORTANT correction needed: The diagram contains the label "TRIM" pointing to the special momentum points on the k-axis, which is a key concept the student is supposed to identify and explain as part of the calculation method for the Z2 invariant.. Specifically: The label "TRIM" points to the Time-. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 4 10.0 points

Examine the diagram below and describe the quantum spin Hall effect and the nature of edge states that arise due to its presence in topological insulators. Highlight the role of spin-momentum locking.

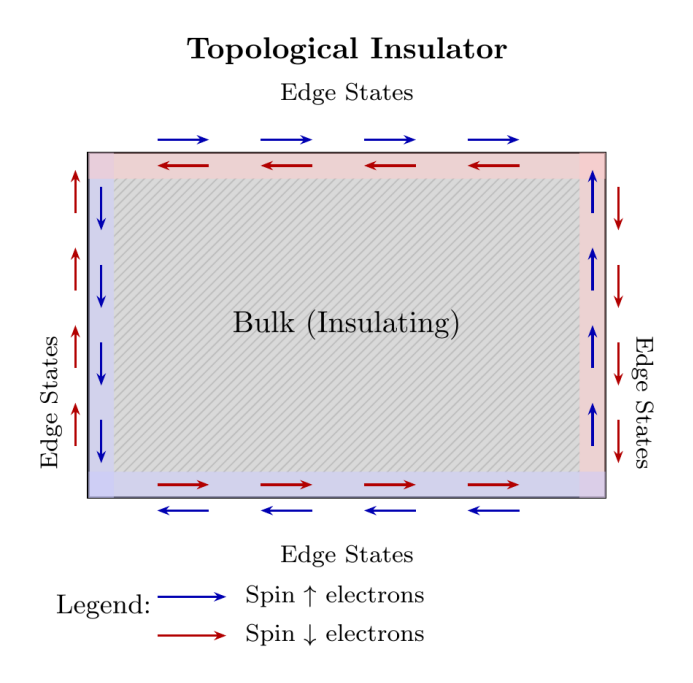

Fig. 4. A diagram of a topological insulator showing the bulk and edge regions. The bulk should be represented as an insulating material, while the edges should show conducting channels. Include arrows along the edges to represent the direction of electron flow, with each arrow labeled to indicate the electron spin direction (up or down) to demonstrate spin-momentum locking. The diagram should clearly differentiate between the insulating bulk and the conducting edge states, highlighting the quantum spin Hall effect. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • spin-momentum locking The diagram will FAIL review if any of these appear.

Question 5

10.0 points

Using the k-space diagram below, identify and explain the regions corresponding to trivial and non-trivial topological phases. Discuss the implications for electronic transport properties.

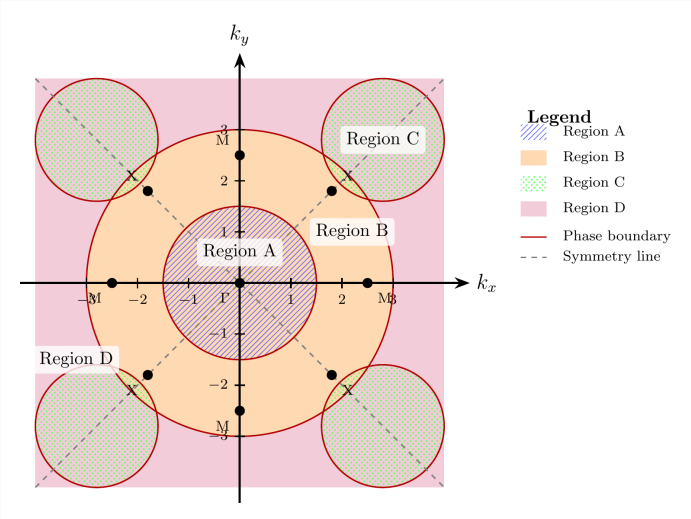

Fig. 5. A k-space diagram illustrating topological phases. The diagram should show a two-dimensional k-space with regions labeled as 'trivial' and 'non-trivial' topological phases. Use different colors or shading to distinguish these regions. Include axes labeled kx and ky, and mark the boundaries between the phases clearly. The diagram should be suitable for discussing electronic transport properties in condensed matter physics.